

CONSTRUCTION OS

PRD_2

ERP Navigation, Role Workspace & Transaction Integrity Addendum

Additive specification to PRD_1. This document formalizes NAV-inspired enterprise patterns without replacing any approved PRD_1 requirement.

Document	Value
Document code	COS-PRD-002
Version	1.0
Status	Developer handover - approved addendum
Date	02 August 2026
Base specification	Construction OS PRD_1
Implementation principle	Backward-compatible and feature-flagged

Document precedence

PRD_1 remains the complete base product specification. PRD_2 adds and clarifies only the features explicitly listed here. Where a requirement is not mentioned in PRD_2, PRD_1 remains unchanged. Where an overlap exists, the compatibility matrix in Section 2 defines the intended interpretation.

01

Document Governance

Version control, scope boundaries and implementation rules for using PRD_1 and PRD_2 together.

1.1 Purpose

PRD_2 converts selected enterprise resource planning patterns into build-ready Construction OS requirements. It focuses on role-specific workspaces, consistent entity navigation, immutable transaction records, operational journals, analytical dimensions, and action-oriented work queues.

1.2 Additive-only rule

- No PRD_1 module, workflow, permission, dashboard, data entity, subscription decision or user role is removed by PRD_2.
- PRD_2 features must be introduced through backward-compatible database migrations and feature flags.
- Existing records retain their identifiers, ownership, permissions, timelines and audit history.
- PRD_1 terminology remains authoritative unless PRD_2 explicitly defines a technical subtype such as Posted, Ledger Entry or Journal Batch.
- Company administrators may enable new features by role, module, branch, company or subscription entitlement where applicable.

1.3 Product decisions retained from PRD_1

Confirmed decision	PRD_2 treatment
Fixed dashboard per user type	Retained. Formalized as a fixed Role Center; no end-user drag-and-drop dashboard builder.
Manual save	Retained. Drafts are saved explicitly; posting is a separate deliberate action.
Universal Draft Mode	Retained and extended with Submit, Approve, Post, Reverse and Archive states.
Every object has a timeline	Retained. Posting, reversals, journals and access requests add timeline events.
Hybrid approval engine	Retained. Posting can require completion of configured sequential/parallel approvals.
Locked global search results	Retained. Restricted results remain visible by title/metadata and support access requests.
Unique permanent object IDs	Retained. Implemented through a centralized Number Series service.
Role-based and individual permissions	Retained and applied to cards, FactBoxes, journals, ledgers and posted records.
Responsive desktop/mobile design	Retained. PRD_2 defines responsive behavior for FactBoxes, lists and work queues.
Offline mode	Retained with restrictions: drafts may be created offline; financial or inventory posting requires server confirmation.

1.4 Out of scope for PRD_2

- Pricing changes, marketplace commissions and professional subscription models.
- Replacement of the approved visual identity or approved dashboard renders.
- A no-code module builder or unrestricted company-created modules.
- Keyboard-first command palettes in V1; PRD_1 selected no keyboard shortcuts for the initial release.
- Automatic graphical drawing comparison, unless separately commissioned.

02

Compatibility Matrix

A direct mapping between the existing product specification and the new enterprise patterns.

PRD_1 capability	PRD_2 pattern	Compatibility rule
Role dashboards	Role Center	Formalizes the fixed dashboard for each user type. Existing dashboard layouts remain the design baseline.
Dashboard KPI cards	Cues	Adds click-through filtering, refresh rules, thresholds and permissions.
360-degree entity profiles	Card Pages	Defines a standard page structure for Project, Contractor, Employee, Contract, Company, Drawing and other entities.
Related Items panel	FactBox / Relationship Panel	Defines the right-side summary panel and mobile drawer behavior. Hybrid automatic/manual linking remains unchanged.
Unique IDs	Number Series	Adds centralized generation, prefixes, annual resets and collision prevention.
Project and cost metadata	Dimensions	Adds reusable analytical dimensions without changing record ownership or permissions.
Universal Draft Mode	Posting lifecycle	Adds immutable finalization after approval for financial, stock and selected legal records.
Stock in/out and daily close	Journals	Adds controlled batch preparation, validation and posting for movements.
Audit logs and timelines	Ledgers	Adds append-only accounting-style entries and reversal links.
Global search	Enterprise search	Adds saved filters, locked-result handling and access-request routing.
Notifications	Work Inbox	Separates actionable items from informational notifications.
Favorites and recent items	Workspace Tabs	Adds role-safe pinned records and recent-context navigation.

Non-interference requirement

Developers must not rename existing customer-facing modules to match ERP terminology. “Role Center,” “FactBox,” “Journal,” and “Ledger” are internal architecture terms. The user interface should continue using Construction OS language such as Dashboard, Related Information, Stock Movements and Activity History.

03

Role Centers and Cues

A fixed, permission-aware front page for each role, optimized around decisions and pending work.

3.1 Role Center definition

A Role Center is the first authenticated screen displayed after a user enters a company workspace. It is assigned by the Company Admin through the user position and role configuration. A user with multiple authorized roles may switch Role Centers, but each Role Center layout remains platform-defined and fixed.

3.2 Standard Role Center regions

Region	Requirement
Header	Company identity, workspace/branch context, global search, notifications, user profile and help.
Primary cues	Four to eight role-critical KPI tiles with count, status, trend and click-through destination.
Work Inbox	Tasks, approvals, signatures, comments or corrections requiring the user's action.
Operational list	Role-specific records such as projects, drawing packages, invoices, employees or stock movements.
Recent activity	Permission-filtered timeline events relevant to the role.
Context panel	Optional upcoming deadlines, risk alerts, meetings or account balances.

3.3 Initial Role Center catalog

Role Center	Required cues	Primary operational content
CEO / Company Owner	Active projects, revenue, pending approvals, risks	Project health, financial overview, approvals, subsidiaries, critical alerts.
Company Admin	Licensed users, pending invitations, role distribution, security alerts	User list, create user, assign role, access mode, permissions and audit activity.
Architect	Active projects, pending drawings, open RFIs, revisions awaiting approval	Drawing packages, approvals, design tasks, submittals and recent activity.
Engineer	Technical tasks, open RFIs, inspections, approvals	Engineering packages, calculations, coordination issues and deadlines.
Project Manager	Project progress, overdue tasks, budget variance, open issues	Milestones, team workload, risks, approvals and site reports.
HR	Employees, absences, leave requests, vacancies	Employee records, attendance exceptions, onboarding and recruitment.
Finance	Revenue, expenses, cash balance, receivables	Transactions, invoices, spending approvals, budgets and upcoming payments.
Legal	Contracts under review, claims, penalties, expiring documents	Contract queue, legal actions, disputes, signatures and compliance.
Stock Manager	Low stock, movements today, pending confirmations, daily close	Warehouse items, stock journal, issued goods, tagged recipients and discrepancies.
Sales	Available units, active leads, reservations, floor-plan requests	CRM pipeline, client requests, sales documents and pending technical responses.
Contractor	Assigned tasks, pending acknowledgements, penalties, invoices	Own contracts, MoM records, submissions, payment status and response queue.
Client	Pending approvals, project progress, variations, requests	Approved project information, decisions, documents and collaboration items.

3.4 Cue behavior

- Each cue displays a current count/value, status label and optional comparison period.

CONSTRUCTION OS / PRODUCT REQUIREMENTS

- Selecting a cue opens the related list with the exact filter already applied.
- Cue calculations must respect record-level visibility, company/JV boundaries and field-level permissions.
- Cue thresholds are platform defaults; authorized administrators may select from approved threshold presets where subscription permits.
- Cue counts must never include records the user cannot open, except separately designed locked-search indicators.
- Cue snapshots should be cached and refreshed after relevant domain events or at a configurable interval.

3.5 Acceptance criteria

- An Architect logging in sees the approved Architect Dashboard and not Finance or HR-only cues.
- Selecting “Pending Drawings: 18” opens a drawing list filtered to pending drawings accessible to that architect.
- A Company Admin can assign or change a user’s Role Center without changing historical task ownership.
- If a user belongs to two companies, each company opens its own branded Role Center and isolated data context.

04

Unified List, Card and FactBox Pattern

A consistent navigation model for every major object, reducing training time and context switching.

4.1 List pages

- Every operational module opens to a searchable, filterable and permission-aware list page.
- List pages support sorting, pagination or virtual scrolling, fixed platform status colors, column selection where permitted, recent filters and export based on permission.
- Users may save personal list views without changing the fixed Role Center layout.
- Company Admins may publish company-standard views for roles or departments.
- Large tables switch to responsive cards on mobile while preserving primary actions and status indicators.

4.2 Card pages (360-degree profiles)

Every master entity must have a dedicated full-page Card rather than relying only on modals. The Card is the authoritative 360-degree view for that record.

Card section	Required content
Header	Permanent ID, name/title, status, owner, company/project context, visibility and primary actions.
Summary	Key facts and role-permitted KPIs.
Tabs	Overview, Timeline, Tasks, Documents, Approvals, Comments, Relationships and module-specific tabs.
FactBox	Compact related facts and shortcuts.
Audit information	Created by/date, modified by/date, current version and posting state where applicable.
Help	Minimal “?” tooltips for labels and fields, consistent with PRD_1.

4.3 Required Card catalog

- Company, parent company, subsidiary, branch and professional profile.
- Project, building, floor, zone, unit and Digital Twin object.
- Employee, user account, department, team, contractor and client.
- Task, decision task, meeting, Minutes of Meeting, RFI, issue and risk.
- Contract, variation, penalty, claim, invoice, expense and payment.
- Drawing, revision, submittal, document and approval request.
- Warehouse, material, asset, equipment, vehicle and stock movement.

4.4 FactBox / Related Information panel

Viewport	Behavior
Desktop	Persistent right-side panel; collapsible; updates when the selected list row changes.
Tablet	Slide-over drawer opened from “Related Information”.
Mobile	Bottom sheet or full-screen detail drawer with grouped sections.

- FactBoxes may show contacts, active contracts, balances, pending approvals, linked documents, project status, contractor score or recent timeline entries.
- FactBox content must be permission-filtered independently from the parent record.
- Selecting a FactBox item navigates to its Card Page and preserves a Back path to the original context.
- FactBoxes are read-mostly; critical edits open the full Card or an approved action form.

05

Work Inbox and Workspace Context

An action-first layer that helps each user start the day with work requiring attention.

5.1 Work Inbox

The Work Inbox is distinct from the notification center. Notifications communicate events; Work Inbox items require or request an action.

Inbox category	Examples	Completion rule
Approval	Drawing, contract, expense, variation, invoice	Approve, reject or request revision.
Signature	Minutes of Meeting, contract, acknowledgement	Sign or formally decline with comment.
Task action	Assigned task, contributor deliverable, overdue explanation	Complete required action or submit mandatory comment.
Review	RFI answer, contractor penalty response, access request	Record a decision or route to another approver.
Data correction	Stock discrepancy, rejected document, invalid expense	Correct and resubmit.
Daily confirmation	Warehouse daily close, attendance exception	Confirm no movement/exception or enter required data.

- Every item has priority, due date, source record, assigned user/role, escalation path and completion status.
- One action may create multiple inbox items for parallel approvers; completion rules follow the approval configuration.
- Inbox items are automatically closed only when the underlying domain action succeeds.
- Overdue inbox items trigger escalation rules and remain visible in audit history.

5.2 Workspace tabs and recent context

- Users may pin a limited number of authorized records such as active projects, contracts or drawings as workspace tabs.
- Tabs are personal navigation state, not shared company data and not billable objects.
- Revoked permissions immediately remove or lock the tab.
- The system maintains Recently Opened records and Favorite Projects as already approved in PRD_1.
- On unexpected browser closure, the next login asks whether to restore the previous workspace, per PRD_1.

06

Master Data and Number Series

Centralized authoritative records and permanent identifiers across all modules and companies.

6.1 Master data principles

- A real-world entity should have one master record per tenant unless a controlled cross-company link is required.
- Duplicate detection is required for companies, contractors, clients, materials, professional profiles and bank accounts.
- Where an external company already exists in the Construction OS network, an internal company should link the network identity rather than create an uncontrolled duplicate.
- Private internal fields, contracts and performance records remain isolated even when a public company identity is linked.

6.2 Number Series service

Entity	Default format	Example
Project	PRJ-{YYYY}-{SEQ6}	PRJ-2026-000154
Task	TSK-{SEQ7}	TSK-0005821
Decision	DEC-{YYYY}-{SEQ5}	DEC-2026-00142
Contract	CON-{YYYY}-{SEQ5}	CON-2026-00321
Contractor	CTR-{SEQ6}	CTR-000093
Meeting / MoM	MOM-{YYYY}-{SEQ5}	MOM-2026-00109
Drawing	DRW-{DISC}-{SEQ6}	DRW-A-004521
RFI	RFI-{YYYY}-{SEQ5}	RFI-2026-00491
Expense	EXP-{YYYY}-{SEQ6}	EXP-2026-001920
Stock movement	STM-{YYYY}-{SEQ7}	STM-2026-0004821
Posting batch	BAT-{TYPE}-{YYYY}-{SEQ5}	BAT-STK-2026-00028

- Identifiers are permanent and cannot be reused after deletion, reversal, cancellation or tenant migration.
- Sequence generation must be atomic and safe across concurrent application servers.
- The system stores both the permanent technical UUID and the human-readable number.
- Company-specific prefixes may be supported as a subscription configuration, but uniqueness must remain guaranteed within the tenant.
- JV records may include the owning company code while keeping one shared JV record identifier.

07

Analytical Dimensions

Multi-axis reporting and filtering without duplicating transactions or changing record permissions.

7.1 Dimension concept

A Dimension is a reusable analytical label attached to a record or ledger entry. Dimensions enable the same expense, stock movement, task or progress entry to be analyzed by multiple business perspectives.

7.2 Initial dimension catalog

Dimension	Example values	Primary use
Project	Blue Villa, Tower A	All operational and financial analysis.
Branch	Tirana HQ, Vlore Branch	Regional responsibility and reporting.
Department	Architecture, Finance, Legal	Department cost and workload.
Cost Center	Design, Site Operations, Head Office	Budget control.
Building / Block	Block A, Hotel Tower	Project breakdown.
Floor / Zone	Level 07, Zone East	Drawings, tasks, stock and progress.
Phase	Design, Structure, Facade, Fit-out	Schedule and cost analysis.
Trade	Facade, HVAC, Electrical	Contractor and procurement analysis.
Contractor	ABC Facades	Cost, quality and performance.
JV Company	Company A, Company B	Shared project contribution reporting.
Funding Source	Bank Loan, Equity	Finance reporting where enabled.

7.3 Rules

- Dimensions are not security permissions. Visibility remains governed by tenant, record and role permissions.
- Required dimensions can be configured by transaction type; e.g., a project expense requires Project and Cost Center.
- Default dimensions may inherit from Project, Department, Contractor, Warehouse or User profile.
- Conflicting default dimensions must trigger a visible validation message before submission or posting.
- Posted ledger dimensions are immutable; corrections require a reversing entry and a corrected repost.
- Reports must support filtering, grouping and pivoting by one or more dimensions.

7.4 Example

Expense: EUR 15,000

Project = Blue Villa | Department = Architecture | Building = Block A | Phase = Facade | Contractor = ABC Facades | Cost Center = Design. One expense is stored once and analyzed through all attached dimensions.

08

Draft, Approval, Posting and Locking

A deliberate separation between editable preparation and immutable business records.

8.1 State model

State	Meaning	Editable?
Draft	Visible only to author and authorized collaborators.	Yes
Submitted	Ready for review; changes require withdrawal or reviewer request.	Restricted
Under Review	Approval workflow active.	No, except comments/attachments allowed by workflow
Approved	Business approval completed but final posting may still be pending.	No
Posted	Committed to ledger and operational balances.	No
Reversed	Original posted entry neutralized by linked reversal.	No
Archived	Closed/read-only and retained for search/export.	No

8.2 Records requiring posting

Module	Examples	Posting effect
Finance	Invoices, payments, expenses, contractor penalties/fees	Updates balances, budget actuals, receivables/payables and financial ledger.
Warehouse	Receipts, issues, returns, transfers, adjustments	Updates on-hand, reserved and project-consumed stock.
HR / Time	Approved timesheets or payroll export batches where enabled	Locks period data and creates payroll/time ledger entries.
Project progress	Certified quantity/progress batch	Updates approved progress and contract milestone metrics.
Legal / Contract	Executed contract or approved variation register entry	Locks signed version and updates contract value/obligations.

8.3 Posting rules

- Posting is a server-side transaction and must be idempotent.
- All required approvals, dimensions, attachments and validation rules must pass before posting.
- The user must have explicit Post permission for the record type and company context.
- Posted records cannot be edited or deleted. Corrections use reversal, credit, return or adjustment flows.
- Posting generates ledger entries, timeline events, notifications, audit entries and balance updates in one atomic transaction.
- If any posting step fails, the entire transaction rolls back and the record remains approved/unposted with a clear error.

8.4 Offline constraint

Posting requires connectivity

Drafts, photos and journal lines may be prepared offline. Final finance, stock, time or progress posting requires online server validation to prevent duplicate or conflicting entries.

09

Operational Journals and Batch Processing

Controlled preparation, review and posting of high-volume operational changes.

9.1 Journal concept

A Journal is a temporary batch of structured lines that allows users to prepare multiple related movements, validate them together and post them as one controlled operation. The customer-facing names should remain familiar, such as Stock Movements, Daily Attendance or Expense Batch.

9.2 Initial journal types

Journal	Typical lines	Posting result
Stock Receipt	Material, quantity, supplier, PO, warehouse, condition	Increases stock and creates receipt entries.
Stock Issue	Material, quantity, project, destination, tagged receiver	Decreases stock and creates recipient confirmation request.
Warehouse Transfer	From warehouse, to warehouse, material, quantity	Dispatch entry plus pending receipt; closes after receiving confirmation.
Stock Adjustment	Counted quantity, system quantity, reason, evidence	Creates variance entry after approval.
Expense Batch	Receipt, amount, project, category, superior	Creates approved expense/payable entries.
Attendance Exception	Employee, date, absence/late reason, supervisor	Updates HR attendance after HR review.
Progress Certification	BOQ item, completed quantity, evidence, approver	Updates certified project/contract progress.

9.3 Journal controls

- Each batch has owner, company, branch, journal type, posting date, status and optional approver.
- Lines can be imported from approved templates or entered manually.
- Validate checks required fields, dimensions, duplicates, balance availability and permissions without posting.
- Post locks the batch and creates a posting result with entry references.
- A failed line blocks the entire batch by default; authorized configurations may allow split posting only for explicitly independent lines.
- The journal must show total debits/credits or total quantities where relevant before posting.

9.4 Warehouse daily close integration

- At the configured end of day, each Stock Manager receives a persistent Work Inbox item.
- The manager selects No Movement or enters/validates all received, issued, transferred, returned, damaged or corrected goods.
- The daily close posts the day's validated stock journal batches or records an explicit no-movement confirmation.
- Missing confirmation escalates to the superior and remains in the audit trail, as required by PRD_1.

10

Immutable Ledgers and Reversals

Append-only records that protect financial, stock, legal and operational history.

10.1 Ledger principles

- A ledger entry is append-only and created only by a successful posting transaction or approved system event.
- Ledger entries are never edited in place and never hard-deleted.
- Every entry references the source record, posting batch, user, timestamp, company, dimensions and related entries.
- Corrections create new entries linked through `ReversesEntryId` or `CorrectsEntryId`.
- Card Pages and reports display the current net effect while preserving the complete entry chain.

10.2 Initial ledger catalog

Ledger	Core entry data
Financial Ledger	Amount, currency, account, debit/credit, tax, project, contractor, dimensions.
Stock Ledger	Material, quantity, unit, warehouse, project, movement type, unit cost, recipient.
Contract Ledger	Base value, variation, penalty, legal fee, retention, certified and paid changes.
Approval Ledger	Approver, decision, sequence/parallel group, comment, timestamp, record version.
Document Ledger	File version, hash, uploader, revision, approval, signature and supersession.
Access Ledger	Permission grant/revoke, access request, support access, duration and approver.
Progress Ledger	Work package, certified quantity/percentage, source evidence, approver and period.
Time Ledger	Employee, date, hours, project, activity, approver and posting batch.

10.3 Reversal workflow

1. Authorized user opens the posted source record and selects Reverse or Correct.
2. System requires a reason, supporting document where configured and any necessary approval.
3. System creates a new draft reversal referencing the original posted entries.
4. After approval, the reversal posts equal and opposite ledger effects.
5. If correction is required, the user creates a new corrected draft and posts it after the reversal.
6. Timeline and reports show Original, Reversal and Correction as a linked chain.

11

Enterprise Lists, Filters and Search

Fast navigation through high-volume construction data while respecting access rules.

11.1 Filtering requirements

- Every list supports multi-field filters, date ranges, status, dimensions, ownership and related-record filters.
- Filters must support AND logic; approved advanced filters may support grouped OR conditions.
- Users can save personal views; admins can publish role-standard views.
- Selecting a cue or FactBox applies a transparent filter chip that users can inspect and clear.
- Filter state is retained when navigating from List to Card and back.
- Exports use the current filter and include only visible/permitted columns.

11.2 Global search behavior

Result type	Accessible result	Restricted result
Title / ID	Shown and clickable.	Shown with lock indicator.
Metadata	Allowed metadata shown.	Only safe metadata: title, type, owning department/company and access-request action.
Content snippets	Shown if indexed and permitted.	Never shown.
Action	Open record.	Request access with mandatory reason.

11.3 Access request flow

7. User selects a locked search result.
8. System explains current restriction and requires a business reason.
9. Request routes to the record owner or configured approver.
10. Approver chooses reject, permanent access, temporary access or read-only access.
11. Grant is applied at record or defined scope and appears in the Access Ledger.
12. Expiration automatically revokes temporary access and closes the grant timeline.

11.4 Search performance

- Search targets: first results under 2 seconds for normal tenant sizes; progressive results may continue loading.
- Indexing must be asynchronous but near real-time after record creation, posting, permission change or deletion/archive.
- Permission changes must invalidate cached search visibility immediately or within a defined maximum of 60 seconds.

12

Personalization Boundaries and Responsive UX

Premium consistency without allowing role dashboards to become fragmented or unsupported.

12.1 Allowed personalization

Allowed	Not allowed in V1
Choose calendar default: day, week or month.	Drag/reorder Role Center widgets.
Select 12/24-hour display, with 12-hour as default.	Change platform status colors.
Choose visible list columns where company policy permits.	Create arbitrary custom modules.
Save list filters and favorite projects.	Replace platform workflow states.
Pin authorized records and restore previous workspace.	Use keyboard shortcut command palette.
Choose notification channels per event type.	Override security, approval or posting rules.

12.2 Responsive behavior

Component	Desktop	Mobile
Role Center	Multi-column dashboard with limited high-value regions.	Single-column priority stack; Work Inbox first.
List	Full table, sticky header, FactBox side panel.	Card list with filter drawer and detail bottom sheet.
Card Page	Header + tabs + FactBox.	Header + segmented tabs; related information in drawer.
Journal	Grid entry and batch totals.	Guided line-entry flow; batch summary before posting.
Approval	Side-by-side document and decision panel.	Document preview followed by fixed decision actions.

12.3 Premium interaction principles

- Flat light interface, dark-grey typography, restrained gold/brand accents and subtle motion.
- Few high-value options in the main workspace; secondary details disclosed through tabs, FactBoxes and drawers.
- Consistent placement of Save, Submit, Approve, Post and More actions across modules.
- Destructive, posting and reversal actions require explicit confirmation and clear consequence text.
- Every unfamiliar label retains the minimal “?” help system specified in PRD_1.

13

Data Model Additions

New entities required to implement PRD_2 without altering the meaning of existing PRD_1 entities.

Entity	Purpose	Scope
RoleCenterDefinition	Defines a fixed dashboard for a role and module entitlement.	Tenant/Platform
RoleCenterAssignment	Links user/company role to Role Center.	Tenant
CueDefinition	Calculation, filter, threshold and destination for a KPI tile.	Platform/Tenant config
CueSnapshot	Cached permission-aware cue value.	Tenant/User
FactBoxDefinition	Defines related information panel sections for an entity type.	Platform
NumberSeries	Prefix, pattern, next value, reset and tenant scope.	Tenant
DimensionDefinition	Dimension code, label, allowed entity types and requirement rules.	Tenant
DimensionValue	Allowed value such as project, department or phase.	Tenant
RecordDimension	Links a record or ledger entry to a dimension value.	Tenant
JournalTemplate	Defines journal type and permitted line schema.	Platform/Tenant
JournalBatch	Header, owner, status, posting date and totals.	Tenant
JournalLine	Prepared movement/transaction line.	Tenant
PostingTransaction	Atomic posting request, idempotency key and result.	Tenant
LedgerEntry	Immutable domain entry with source and dimensions.	Tenant
LedgerLink	Reversal, correction, settlement or application relationship.	Tenant
WorkInboxItem	Actionable user/role assignment linked to source record.	Tenant/User
SavedView	Personal or published list filter/column definition.	Tenant/User
WorkspaceTab	Pinned authorized record and UI context.	User
MasterRecordLink	Links internal master data to a network identity or duplicate candidate.	Tenant/Network

13.1 Required common fields

- id (UUID), tenant_id, owning_company_id, created_at, created_by, updated_at, updated_by.
- status, version, row_version/ETag, archived_at and source_record_id where relevant.
- visibility_policy_id and JV sharing metadata for objects that participate in cross-company workspaces.
- posted_at, posted_by, posting_transaction_id and immutable hash for posted/ledger records.

13.2 Integrity constraints

- Unique number series value per tenant/entity scope.
- No update or delete operation allowed on LedgerEntry after insert.
- Posted source records require matching posting transaction and at least one ledger entry where applicable.
- Reversal links cannot form cycles and must reference entries in the same tenant and compatible domain.
- RecordDimension values must belong to the same tenant and be valid for the record type.

14

API and Event Requirements

Service contracts and domain events required for scalable frontend, backend and integrations.

14.1 API groups

API group	Purpose
/role-centers	Resolve current role workspace, cues, inbox and layout.
/cues	Retrieve/recalculate cue values and drill-down filters.
/entities/{type}	List and Card access using common filtering and pagination conventions.
/number-series	Preview and allocate human-readable identifiers.
/dimensions	Manage definitions, values, defaults and record assignments.
/journals	Create batches, add/import lines, validate and submit.
/posting	Post approved batches/records with idempotency key.
/ledgers	Read ledger entries and linked reversal/correction chains.
/work-inbox	List, assign, complete, delegate and escalate action items.
/saved-views	Save personal views and publish approved role views.
/workspace-tabs	Pin, reorder and close authorized context tabs.
/search	Global search, locked-result metadata and access request initiation.
/access-requests	Approve/reject/expire record access grants.

14.2 Required domain events

- RecordCreated, RecordSubmitted, RecordApproved, RecordRejected, RecordPosted, RecordReversed, RecordArchived
- JournalValidated, JournalPosted, PostingFailed
- LedgerEntryCreated, BalanceChanged, StockAvailabilityChanged
- CueInvalidated, WorkInboxItemCreated, WorkInboxItemCompleted, WorkInboxItemEscalated
- AccessRequested, AccessGranted, AccessExpired, AccessRevoked
- DimensionAssigned, NumberAllocated, MasterRecordLinked

14.3 Idempotency and concurrency

- All posting endpoints require an idempotency key and return the original result for duplicate retries.
- Editable drafts use optimistic concurrency through row version/ETag.
- Number allocation and ledger writes use database transactions or equivalent distributed consistency controls.
- Offline synchronization must detect stale draft versions and require merge/review rather than silently overwrite.

15

Security, Permissions and Separation of Duties

Enterprise controls applied to role navigation, posting and sensitive information.

15.1 New permission actions

Permission	Description
View Role Center	Open the assigned dashboard in a company context.
Manage Role Centers	Assign platform-defined Role Centers to users/roles.
View FactBox	View a specific related-information section.
Manage Dimensions	Create or change tenant dimension definitions/values.
Prepare Journal	Create and edit draft journal batches/lines.
Validate Journal	Run controlled validation.
Approve Journal	Approve a journal where workflow requires.
Post Transaction	Commit approved records and create ledger entries.
Reverse Transaction	Initiate reversal/correction workflows.
View Ledger	Read domain ledgers within permitted scope.
Publish Saved View	Create a company-standard role view.

15.2 Separation-of-duties rules

- A company may configure that the same user cannot both prepare and post a financial or high-value stock batch.
- Threshold-based approvals remain controlled by PRD_1 approval chains.
- Company Admin role assignment does not automatically grant Finance posting or Legal signature permissions.
- Platform Main Admin has no default access to tenant transactions; temporary support access remains governed by PRD_1.
- Locked FactBox sections and Card tabs must not leak totals, counts or snippets without permission.

15.3 Audit requirements

- Log Role Center assignment changes, saved-view publication, dimension changes, posting attempts, reversals and access grants.
- Audit entries include actor, company, device/session, timestamp, source IP where legally permitted, old/new values and reason.
- Audit logs remain Admin-only as selected in PRD_1, with platform access limited to approved support sessions.

16

Performance, Reliability and Offline Behavior

Measurable quality targets for the new enterprise capabilities.

16.1 Performance targets

Operation	Target
Role Center initial load	Primary layout and cached cues within 2.5 seconds at p95 under normal load.
Cue drill-down	Filtered list first page within 2 seconds at p95.
List filtering	Server response within 2 seconds for indexed filters.
Card Page	Core record and summary within 2 seconds; secondary tabs lazy-loaded.
Posting	Typical single/batch post within 5 seconds; long operations use progress state and safe retry.
Global search	First result set within 2 seconds for normal tenant sizes.
Permission revocation	Effective immediately for API authorization; UI/cache invalidation within 60 seconds maximum.

16.2 Reliability

- Posting, ledger creation, balance updates and timeline events must be atomic or use a transactional outbox with guaranteed reconciliation.
- Failed background cue/index updates must retry without blocking core record operations.
- Ledger totals require scheduled reconciliation checks and alerting for inconsistencies.
- Daily backups and disaster recovery follow PRD_1 infrastructure requirements; new ledger tables receive the same or stronger protection.

16.3 Offline behavior matrix

Action	Offline	On reconnect
View downloaded list/card data	Yes	Refresh and flag changed records.
Create/edit draft	Yes	Sync with conflict detection.
Add comment/photo	Yes	Upload and preserve local timestamp + server receipt timestamp.
Prepare journal lines	Yes	Validate against current server balances.
Approve/sign	Only where cached policy permits	Revalidate authorization and record version.
Post/reverse	No	User must explicitly post after online validation.

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Migration and Rollout

A safe implementation sequence that preserves PRD_1 data and allows controlled adoption.

17.1 Migration rules

- Do not rename or regenerate existing permanent IDs.
- Initialize Number Series counters above the highest existing human-readable ID.
- Existing approved financial/stock records that already changed balances should be migrated as posted legacy entries with traceable source references.
- Existing drafts remain drafts; existing approved-but-not-applied records remain approved/unposted.
- Existing timeline and audit entries remain unchanged; migration creates one system migration event per affected record or batch, not fabricated user actions.
- Dimension defaults may be inferred only from explicit existing relations; ambiguous values remain unassigned and appear in a migration exception report.

17.2 Feature flags

Flag	Default rollout
role_centers_v2	Enabled after dashboard mapping verification.
factbox_framework	Enabled module by module.
number_series_service	Enabled for new records; existing IDs preserved.
dimensions	Enabled first for Finance/Stock, then all supported modules.
posting_and_ledgers	Pilot tenant, reconciliation period, then general availability.
work_inbox	Parallel run with existing notifications before making it the primary action queue.
saved_views	Enabled after permission/export tests.

17.3 Recommended implementation phases

Phase	Scope	Dependency
A - Navigation foundation	Common List/Card pages, FactBox framework, Role Centers and cues.	PRD_1 roles, dashboards and permissions.
B - Master data	Number Series, duplicate detection, master linking and dimensions.	Phase A common entity framework.
C - Transaction integrity	Journals, posting service, ledgers, reversals and reconciliation.	Phase B dimensions and IDs.
D - Action layer	Work Inbox, access requests, saved views and workspace tabs.	Phase A navigation + PRD_1 notifications.
E - Optimization	Caching, advanced reports, migration tooling and performance hardening.	All previous phases.

Acceptance Criteria

Testable outcomes for design, frontend, backend, QA and security teams.

ID	Acceptance criterion
AC-01	A user receives the correct fixed Role Center from their active company role and cannot access unauthorized role dashboards.
AC-02	A cue count equals the number of records returned by its drill-down filter for that user.
AC-03	Opening a Contractor Card displays only contractor fields, balances and FactBoxes permitted to the current user.
AC-04	Desktop FactBox becomes a drawer/bottom sheet on tablet/mobile without losing linked actions.
AC-05	Existing PRD_1 record IDs remain unchanged after Number Series deployment.
AC-06	Two concurrent record creations cannot receive the same human-readable number.
AC-07	Required dimensions block submission/posting with clear field-specific errors.
AC-08	An approved expense is not reflected in posted balances until an authorized Post action succeeds.
AC-09	A successful posting creates source status, ledger entries, timeline event and audit record atomically.
AC-10	A posted ledger entry cannot be edited or deleted through UI, API or admin tools.
AC-11	Reversal creates equal/opposite entries and retains a visible link to the original.
AC-12	A Stock Issue journal cannot post more quantity than available unless an authorized override workflow explicitly allows it.
AC-13	Tagged stock recipients receive confirmation/dispute inbox items after posting.
AC-14	Warehouse daily close cannot complete while unvalidated movement lines remain.
AC-15	A user can save a personal filtered view without changing another user's view or the fixed dashboard.
AC-16	A restricted search result shows safe metadata only and never exposes content snippets.
AC-17	Temporary access automatically expires and is recorded in the Access Ledger.
AC-18	Work Inbox completion requires successful completion of the source action, not merely clicking dismiss.
AC-19	Offline drafts sync with conflict detection; posting remains unavailable offline.
AC-20	All new pages meet the responsive, accessibility and help-tooltip standards of PRD_1.
AC-21	JV users see only records explicitly shared to their company or selected members.
AC-22	Company Admin can assign an Architect Role Center and credentials without granting Finance posting rights.
AC-23	Role Center cue and list data update after relevant domain events without a manual full-page refresh.
AC-24	Archived projects and their ledger history remain searchable and downloadable according to PRD_1 permissions.

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QA and Test Scenarios

Minimum end-to-end scenarios required before PRD_2 features can be released.

19.1 Critical test packs

Test pack	Required scenarios
Tenant isolation	Role Center, search, FactBox, journal and ledger data never cross tenant boundaries.
JV sharing	Shared records visible only to configured partner companies/members; private records remain isolated.
Posting integrity	Success, rollback, duplicate retry, timeout retry and reconciliation.
Permission changes	Grant, revoke, temporary expiry, cached screen invalidation and offline reconnect.
Number Series	Concurrency, annual reset, tenant prefix, cancelled number and migration continuity.
Dimensions	Defaults, conflict, required values, report filtering and immutable posted values.
Responsive UI	Desktop, tablet and mobile List/Card/FactBox/Journal/Approval workflows.
Offline sync	Draft merge, attachment retry, stale version and forbidden offline posting.
Performance	High-volume list, search index, cue refresh and batch posting.

19.2 Security tests

- Direct API request to unauthorized FactBox or ledger endpoint returns forbidden without metadata leakage.
- Role Center URL manipulation cannot open another role's data.
- Posting permission is checked server-side and cannot be bypassed through frontend state.
- Restricted search results never include document body, amount, legal detail or private comment unless specifically allowed.
- Support access expiration terminates active sessions or removes tenant access immediately according to policy.

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Developer Handover Checklist

Artifacts the senior full-stack team must produce before implementation approval.

Deliverable	Expected output
Architecture decision records	Service boundaries for Role Center, search, posting, ledger, dimensions and offline sync.
Database design	ERD, indexes, constraints, partitioning/tenant strategy and migration scripts.
API contract	OpenAPI specification with authorization, errors, pagination and idempotency.
Frontend component map	Reusable List, Card, FactBox, Cue, Work Inbox and Journal components.
Permission matrix update	New actions mapped to every PRD_1 role.
Event catalog	Domain event schemas, transactional outbox strategy and consumers.
Reconciliation design	Scheduled checks for financial, stock and progress ledger consistency.
Migration plan	Dry-run, exception report, rollback and tenant rollout procedure.
Test plan	Automated unit, integration, E2E, permission, security, offline and performance tests.
Estimation	Team composition, implementation hours, infrastructure impact, risks and phased cost.

20.1 Required technical decisions before coding

- Modular monolith versus service separation for Posting/Ledger/Search, including expected scale thresholds.
- Database transaction and tenant isolation strategy for shared and dedicated database plans.
- Search engine and permission-filtering architecture.
- Offline local database, encryption and synchronization protocol.
- Ledger partitioning, reconciliation and retention strategy.
- Cue caching and invalidation strategy.
- Document hash/signature implementation for posted documents.

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Final Implementation Statement

The intended outcome and the boundary between PRD_1 and PRD_2.

PRD_1 defines the complete Construction OS product, business model, departments, modules, workflows, portals, network, Digital Twin, offline capability, dashboards and design direction. PRD_2 adds the enterprise interaction and transaction patterns required to make that platform consistent, auditable and scalable for daily company operations.

Build principle

The user should experience one premium Construction OS built specifically for their company. The architecture may use ERP-grade Role Centers, Cards, FactBoxes, Journals, Dimensions and Ledgers, but the interface must remain simple, construction-specific and consistent with the approved PRD_1 renders.

21.1 Definition of done

- PRD_2 capabilities are implemented without regression to any PRD_1 feature or permission.
- Existing tenants can migrate without losing identifiers, history, documents or balances.
- Every critical transaction is traceable from source draft through approval, posting, ledger and reversal.
- Every role starts from a focused workspace and reaches related information without unnecessary navigation.
- The platform remains responsive, permission-aware, offline-capable where appropriate and visually premium.

END OF PRD_2